



UNIVERSITY OF MAKATI
J.P. Rizal Extension, West Rembo Makati City

College of Computing and Information Sciences

V CLOUD

Virtualization and Cloud Computing Technologies

Laboratory

Quiz

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Submitted to:

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I. PROBLEM

1. Design a LAN set-up that would simulate the following roles.
 - A. VLAN and Inter VLAN (use computer)
 - B. VLAN and Inter VLAN (use computer and IP Phone)
 - C. Configure server for DHCP, DNS, FTP, and Web Server
 - D. Configure the ACL

II. OUTPUT. <submit your packet tracer file>

<Logical Layout>

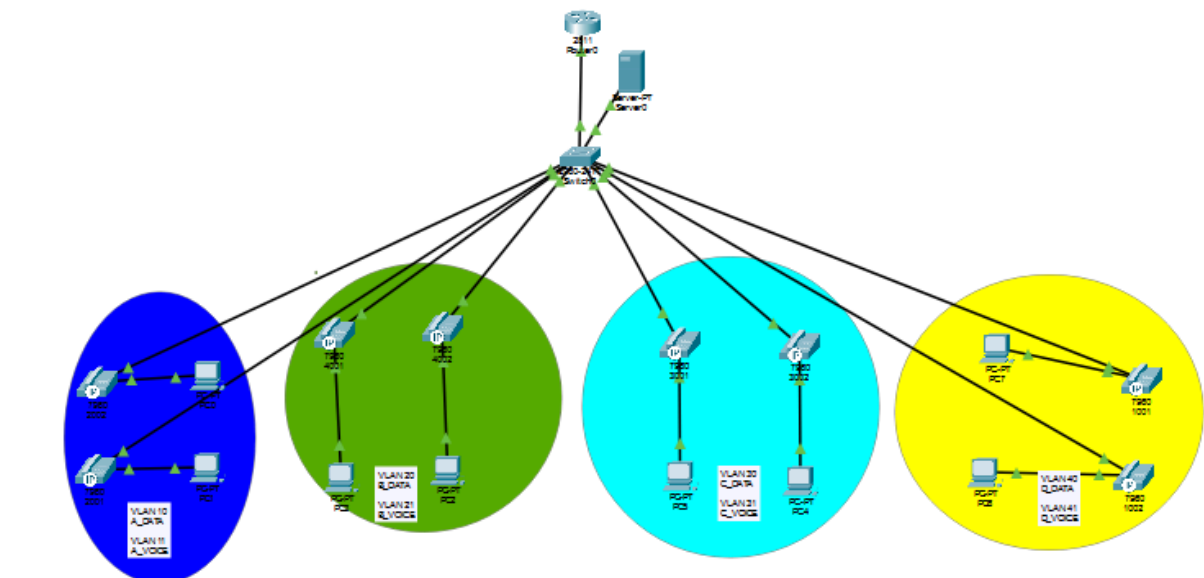
Core Infrastructure

- Router: Cisco 2811 (Handles Inter-VLAN Routing, Access Control Lists, and Call Manager Express).
- Switch: Cisco 2960 24-Port
- Central Server: 192.168.100.10 (Hosts centralized DHCP, DNS, FTP, and Web Services).

VLAN and IP Addressing

- Server Farm | VLAN 100 | 192.168.100.0/24
- Department A | VLAN 10 | 192.168.10.0/24 | VLAN 11 192.168.11.0/24
- Department B | VLAN 20 | 192.168.20.0/24 | VLAN 21 | 192.168.21.0/24
- Department C | VLAN 30 | 192.168.30.0/24 | VLAN 31 | 192.168.31.0/24
- Department D | VLAN 40 | 192.168.40.0/24 | VLAN 41 | 192.168.41.0/24

<Screen shot with discussion>



The network is segmented into separate logical groups to improve security, performance, and traffic management. Each department is assigned its own network:

- Department A – Network 10
- Department B – Network 20
- Department C – Network 30
- Department D – Network 40
- Server Network – 100

This separation isolates broadcast domains, preventing unnecessary traffic between departments and reducing the risk of unauthorized access. Data devices (PCs) and IP phones can also be logically separated to maintain call quality and prioritize voice traffic.

Inter-network communication is handled by the router through trunk links, allowing controlled communication between departments when necessary. The Server Network (100) hosts centralized services such as DHCP, file storage, and internal web systems, ensuring all departments can securely access shared resources while maintaining proper network segmentation.

CLI

```
en
conf t
```

```
vlan 100
name SERVER
vlan 10
name SET_A_DATA
vlan 11
name SET_A_VOICE
vlan 20
name SET_B_DATA
vlan 21
name SET_B_VOICE
vlan 30
name SET_C_DATA
vlan 31
name SET_C_VOICE
vlan 40
name SET_D_DATA
vlan 41
name SET_D_VOICE
```

```
interface f0/1
switchport mode trunk
```

interface f0/2
switchport mode access
switchport access vlan 100

interface range f0/3-4
switchport mode access
switchport access vlan 10
switchport voice vlan 11

interface range f0/5-6
switchport mode access
switchport access vlan 20
switchport voice vlan 21

interface range f0/7-8
switchport mode access
switchport access vlan 30
switchport voice vlan 31

interface range f0/9-10
switchport mode access
switchport access vlan 40
switchport voice vlan 41

enable
configure terminal
interface f0/0
no shutdown

interface f0/0.100
encapsulation dot1Q 100
ip address 192.168.100.1 255.255.255.0

interface f0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
ip helper-address 192.168.100.10

interface f0/0.11
encapsulation dot1Q 11
ip address 192.168.11.1 255.255.255.0
ip helper-address 192.168.100.10

interface f0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.100.10

interface f0/0.21
encapsulation dot1Q 21
ip address 192.168.21.1 255.255.255.0
ip helper-address 192.168.100.10

interface f0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
ip helper-address 192.168.100.10

interface f0/0.31
encapsulation dot1Q 31
ip address 192.168.31.1 255.255.255.0
ip helper-address 192.168.100.10

interface f0/0.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
ip helper-address 192.168.100.10

interface f0/0.41
encapsulation dot1Q 41
ip address 192.168.41.1 255.255.255.0
ip helper-address 192.168.100.10

telephony-service
max-ephones 8
max-dn 8
ip source-address 192.168.100.1 port 2000
auto assign 1 to 8

ephone-dn 1
number 1001
ephone-dn 2
number 1002
ephone-dn 3
number 2001
ephone-dn 4
number 2002
ephone-dn 5
number 3001
ephone-dn 6
number 3002
ephone-dn 7
number 4001
ephone-dn 8
number 4002

! --- SECURITY ACCESS CONTROL LISTS ---

access-list 105 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 105 permit ip any any

access-list 106 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 106 permit ip any any

interface f0/0.10

ip access-group 105 in

interface f0/0.30

ip access-group 106 in